

Bagging and Random Forests

Turn unstable trees into a stable ensemble

Ensemble Methods · Foundations of Machine Learning



Bootstrap Sampling

Sample with replacement — Draw n observations from n rows; repeats are expected.

Out-of-bag rows — About 36.8% of the original rows are absent from one bootstrap sample.

Why it matters — One dataset can imitate many plausible training sets.

$$\left(1 - \frac{1}{n}\right)^n \rightarrow e^{-1} \approx 0.368$$

Worked example: with $n = 1\{, \}000$ rows, each bootstrap sample still has 1,000 draws, but roughly $0.368 \times 1\{, \}000 \approx 368$ rows never get drawn at all — those are that tree's out-of-bag rows.

Bagging = Bootstrap + Aggregate

Train in parallel — Fit the same high-variance learner on many bootstrap samples.

Combine — Average for regression; majority vote for classification.

Effect — Variance falls while the base learner's bias changes little.

1

Dataset

2

Bootstrap samples

3

Deep trees

4

Average / vote

$$\hat{f}_{\text{bag}}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}_b(x)$$

Why Averaging Reduces Variance

Assumption — Treat the B trees as identically distributed with variance σ^2 each.

If independent — Variance of the average shrinks by a full factor of B .

The catch — Real bootstrap trees are correlated, not independent — the next deck derives the floor this creates.

$$\begin{aligned}\text{Var}\left(\frac{1}{B} \sum_{b=1}^B \hat{f}_b(x)\right) &= \frac{1}{B^2} \sum_{b=1}^B \text{Var}(\hat{f}_b(x)) \\ &= \frac{1}{B^2} \cdot B\sigma^2 = \frac{\sigma^2}{B}\end{aligned}$$

Worked example: a single deep tree has prediction variance $\sigma^2 = 4$. Average $B = 25$ independent trees and the variance drops to $4/25 = 0.16$ — a $25\times$ reduction, with the mean (the bias) essentially unchanged.

Why Trees Are the Natural Base Learner

Deep decision tree

Low bias, high variance: plenty of instability for averaging to remove.

Linear regression

Already comparatively stable: bagging has little variance to cancel.

Design rule

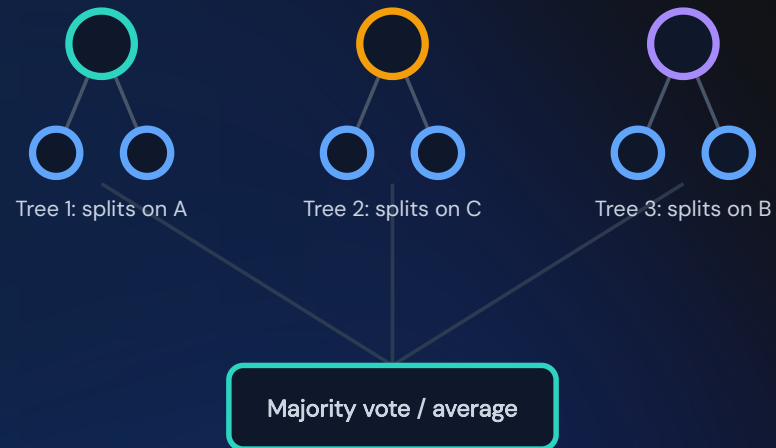
Bagging pays off when base learners are accurate but unstable.

Random Forests Add Feature Randomness

Correlated trees limit averaging — A dominant feature can force nearly every bagged tree into the same early splits.

Random feature subsets — At each split, consider only m of the d available features.

More diversity — Different roots and branches reduce correlation between tree errors.



$m \approx \sqrt{d}$ for classification (a common default)

Out-of-Bag Evaluation

For each row

Predict using only trees whose bootstrap sample omitted that row.

Nearly free validation

Aggregate those predictions into an OOB score without a separate split.

```
from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier(
    n_estimators=200, max_features="sqrt",
    oob_score=True, random_state=0,
)

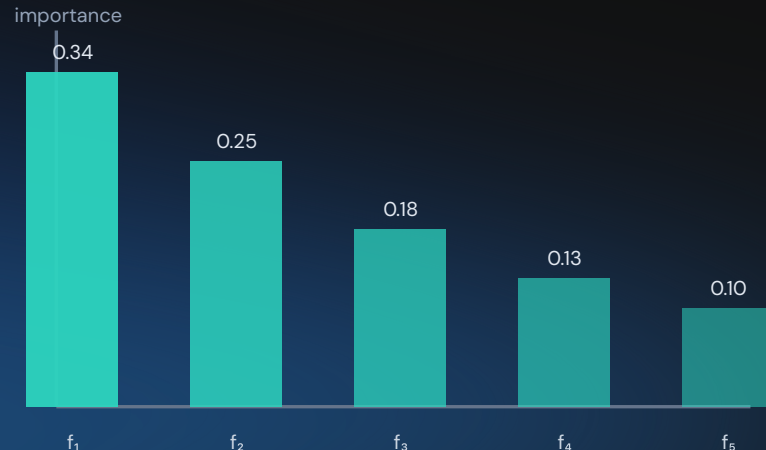
rf.fit(X, y)
print(rf.oob_score_)
```


Feature Importance — Useful, Not Causal

Impurity reduction — Aggregate how much each feature improves splits across all trees.

Practical use — A quick global view for interpretation and feature screening.

Caveats — High-cardinality and correlated features can distort the ranking.



Bagging vs. the Next Idea

Bootstrap

Simulate many training sets.

Bagging

Average high-variance learners.

Random forest

Decorrelate trees with feature randomness.

Next: boosting trains weak learners sequentially, each correcting what remains wrong.